

Pupil-Based Prediction of Affect in VR: A Machine Learning Approach

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ABSTRACT

Adapting VR scenarios in real time based on users' emotional responses is a promising advancement for developers. However, most VR environments are either "blind" to emotions or depend on intrusive, contact-based methods. This study introduces a novel, non-invasive approach for predicting arousal and valence using pupil reactivity and gaze features. Data from 105 participants (ages 18–30) viewing 120 emotional images via a VR headset with eye tracking informed machine learning models. These models, leveraging features like mean and standard deviation of pupil data, achieved F1 scores of 0.64 (valence) and 0.73 (arousal), highlighting potential real-world applications in VR development.

Index Terms: Virtual reality, psychology, pupil reactivity, affective computing, valence, arousal.

1 INTRODUCTION

Algorithmic emotion detection has become a primary interest in the field of human-computer interaction (HCI) [10]. Real-time adaptation of VR scenarios based on emotional reactivity would significantly enhance the capabilities of VR developers. Here, we employ rigorous methodology involving multiple repetitions of standardized stimuli to obtain reliable physiological responses [13]. Our work lays the foundation for a future project, where participants' eye-tracking and pupil reactivity will be recorded during interactions with a 3D virtual environment.

2 RELATED WORKS

Affective states can be inferred from questionnaires, observable behaviors, such as bodily movements [12], facial expression [3] and voice patterns [15]. However, questionnaires validity is limited by subjectivity and systematic errors, e.g. experimenter bias [5], while VR equipment can restrict visibility of facial expressions, and motion tracking may not capture subtle gestures accurately. The user's feedback should be captured using methods that do not interrupt the VR experience, such as physiology [6]. Pupil dilation is thought to be a crucial physiological characteristic for identifying stress and emotion [1] and has already been used for affect prediction in VR [11]. However, despite its potential to provide valuable information, pupil reactivity is infrequently analyzed in usability studies and most studies tend to integrate additional sensor modalities like electroencephalography (EEG), electrodermal response (EDA) or electrocardiography (ECG) [16]. We present a method of arousal and valence prediction based mainly on pupil reactivity and gaze patterns, measured in a completely unobtrusive way.

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3 PROCEDURE

The study involved 105 participants: 50 men and 55 women, aged 18–30 years. We used 120 stimuli from emotional pictures databases: IAPS [8], NAPS [9], GAPED [4], and EmoPics [14], evenly divided between positive, negative, and neutral. Such pictures, although static and 2D, have been chosen due to their well-established norms on valence and arousal, the primary dimensions accounting for significant variance in emotional experiences and their psychophysiological correlates [2]. All the pictures have been equated for the luminance and contrast using MATLAB Shine Toolbox. We used HTC Vive Pro headset, equipped with a dedicated 200Hz Pupil Labs eye-tracking system and external light sensor to record the luminous flux. Each image was presented for 5 seconds, followed by a 4–6 second inter-trial interval with a fixation cross, resulting in a total study duration of 20 minutes. Regular recalibration breaks were implemented to ensure the accuracy of eye-tracking throughout the session and to allow participants to rest, reducing the potential impact of eye fatigue on the data. Pupil Capture software did real-time recording of eye-tracking data while Unity provided session runtime timestamps and trigger tags. After a session, raw data was preprocessed and exported in .csv files using Pupil Player, whereby eye-tracking metrics were combined with session-specific triggers for subsequent detailed analysis.

4 ANALYSIS

Data was pre-processed to remove samples outside of a predefined feasible range (1.5–9 mm diameter), dilation speed outliers, blinking artifacts, trend-line deviation outliers, and temporally isolated samples, using a pupil analysis tool in MATLAB [7], then divided into segments corresponding to each image presentation. Statistical features such as mean, median, kurtosis, standard deviation, and regression coefficients were extracted from these segments through feature engineering, then used to classify valence and arousal levels into high and low categories, as well as for multiclass classification (9 levels) and regression tasks. For binary valence classification, the focus was on distinguishing between extreme values, given the near-normal distribution of labels. 80% of the data were randomly selected for training, and the rest was used for testing.

5 RESULTS

We tested a variety of classifiers with Python and pycaret library achieving the best results using Light Gradient-Boosting Machine. In Table 1, we present a comparison of the F1 score, an evaluation metric that combines precision and recall to measure the accuracy of a specific machine learning classifier. The highest accuracy was achieved in binary classification, with F1 scores of 0.73 for arousal and 0.64 for valence. The multiclass classification was less accurate, yielding F1 scores of 0.31 for arousal and 0.28 for valence. By performing explainability analyses with Shapley Additive Explanations (SHAP) values, we established most influential features for prediction accuracy in binary classification of arousal, i.e. mean of

pupil position, luminous flux, and variance of pupil position. Details can be accessed in supplemental materials.

Table 1: Binary Classification - Arousal and Valence

CLASSIFIER	F1 (Arousal)	F1 (Valence)
Light Gradient Boosting Machine	0.7291	0.6412
Extra Trees Classifier	0.7034	0.6306
Random Forest Classifier	0.6937	0.6263
Gradient Boosting Classifier	0.6737	0.5814
Decision Tree Classifier	0.6286	0.5829
Ada Boost Classifier	0.6240	0.5466
Ridge Classifier	0.5951	0.5520
Linear Discriminant Analysis	0.5929	0.5523
Logistic Regression	0.5569	0.5489
K Neighbors Classifier	0.5243	0.5149
Quadratic Discriminant Analysis	0.5209	-
SVM - Linear Kernel	0.5171	0.5009
Naive Bayes	0.5049	-

6 DISCUSSION

Using machine learning techniques, we predicted the image classification on valence and arousal dimensions with satisfactory precision, with potential for further improvement. Our approach showed worse precision for multiclass predictions, which could be attributed to variability in pupil reactivity, limitations in the feature set or eye fatigue. Future work should focus on exploring deep learning techniques and employing a wider variety of features to improve classification accuracy. Since there is limited research applying these methods to non-contact measures like pupil reactivity in VR, this work represents a pioneering step in the field. Future studies can also concentrate on using dynamic and interactive stimuli to enhance immersion and prevent habituation. Such dynamic setups are particularly valuable as they simulate real-world conditions more effectively and account for the influence of active engagement on emotional states. Examining the function of interaction might also greatly enhance the data's ecological validity.

7 CONCLUSION

We demonstrated that, based on the position and size of the pupil, it is possible to obtain individualized arousal metrics for a specific user in response to a specific stimulus. The pupil position features, mean and variance in particular, as well as luminous flux, are the most impactful features in this model, which align with their relevance in tasks involving visual attention, fixation, and pupil dynamics. These features likely capture core aspects of the participant's interaction with the VR environment. This could lead to the development of simulations sensitive to users' emotional reactivity. Our approach, designed for usage in VR environments, is unobtrusive for the participants, and eye trackers able to record pupil data are already mounted in many VR headsets. Our solution shows promise for industries like gaming, healthcare, and market research, enabling personalized, immersive VR experiences. It supports real-time adaptation, enhancing VR therapy, gaming immersion, and tailored training programs. Moreover, future developments in ML may enhance the accuracy and applicability of this approach, further expanding its utility.

SUPPLEMENTAL MATERIALS

All supplemental materials are available at https://osf.io/txvhk/?view_only=91cc62b084144ed2b796079b061dce83, released under a CC BY 4.0 license. These materials include examples of the images used in the procedure, procedural steps presented in a schematic visual format, the experimental display, Shapley Additive Explanations (SHAP) values and F1 scores for binary as well as multiclass classification.

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REFERENCES

- [1] A. Babiker, I. Faye, K. Prehn, and A. S. Malik. Machine learning to differentiate between positive and negative emotions using pupil diameter. *Frontiers in Psychology*, 6, 2015. 1
- [2] L. P. J. Bradley M. M. Measuring emotion: the self-assessment manikin and the semantic differential. *J. Behav. Ther. Exp. Psychiatry*, 25:49–59, 1994. 1
- [3] E. A. Clark, J. Kessinger, S. E. Duncan, M. A. Bell, J. Lahne, D. L. Gallagher, and S. F. O'Keefe. The facial action coding system for characterization of human affective response to consumer product-based stimuli: a systematic review. *Frontiers in psychology*, 11:920, 2020. 1
- [4] E. S. Dan-Glauser and K. R. Scherer. The geneva affective picture database (gaped): a new 730-picture database focusing on valence and normative significance. *Behavior research methods*, 43:468–477, 2011. 1
- [5] R. J. Fisher. Social desirability bias and the validity of indirect questioning. *Journal of consumer research*, 20(2):303–315, 1993. 1
- [6] T. Georgiou and Y. Demiris. Adaptive user modelling in car racing games using behavioural and physiological data. *User Modeling and User-Adapted Interaction*, 27:267–311, 2017. 1
- [7] M. E. Kret and E. E. Sjak-Shie. Preprocessing pupil size data: Guidelines and code. *Behavior research methods*, 51:1336–1342, 2019. 1
- [8] P. J. Lang, M. M. Bradley, B. N. Cuthbert, et al. International affective picture system (iaps): Technical manual and affective ratings. *NIMH Center for the Study of Emotion and Attention*, 1(39-58):3, 1997. 1
- [9] A. Marchewka, Ł. Żurawski, K. Jednoróg, and A. Grabowska. The nencki affective picture system (naps): Introduction to a novel, standardized, wide-range, high-quality, realistic picture database. *Behavior research methods*, 46:596–610, 2014. 1
- [10] J. Marín-Morales, C. Llinares, J. Guixeres, and M. Alcañiz. Emotion recognition in immersive virtual reality: From statistics to affective computing. *Sensors (Basel, Switzerland)*, 20, 2020. 1
- [11] P. Ren, A. Barreto, Y. Gao, and M. Adjouadi. Affective assessment of computer users based on processing the pupil diameter signal. In *2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, pp. 2594–2597. IEEE, 2011. 1
- [12] P. Robitaille and M. J. McGuffin. Increased affect-arousal in vr can be detected from faster body motion with increased heart rate. In *Proceedings of the ACM SIGGRAPH symposium on interactive 3D graphics and games*, pp. 1–6, 2019. 1
- [13] R. Sterna, A. M. Siry, J. Pilarczyk, and M. J. Kuniecki. Psychophysiology in studying vr-mediated interactions: Panacea or a trick? valuable applications, limitations, and future directions. *Frontiers in Virtual Reality*, 2:648318, 2021. 1
- [14] M. Wessa, P. Kanske, P. Neumeister, K. Bode, J. Heissler, S. Schönfelder, et al. Emopics: Subjektive und psychophysiologische evaluation neuen bildmaterials für die klinisch-bio-psychologische forschung. *Zeitschrift für Klinische Psychologie und Psychotherapie*, 39(Suppl. 1/11):77, 2010. 1
- [15] J. R. Williamson, T. F. Quatieri, B. S. Helfer, G. Ciccarelli, and D. D. Mehta. Vocal and facial biomarkers of depression based on motor incoordination and timing. In *Proceedings of the 4th international workshop on audio/visual emotion challenge*, pp. 65–72, 2014. 1
- [16] L. Zhang, J. Wade, D. Bian, J. Fan, A. Swanson, A. Weitlauf, Z. Warren, and N. Sarkar. Cognitive load measurement in a virtual reality-based driving system for autism intervention. *IEEE Transactions on Affective Computing*, 8(2):176–189, 2017. 1